

8.7 Triangularization

Section 4 found two ways for diagonalization to fail. The field can be too small, as for a rotation of $\mathbb{R}^2$. Or there can be too few eigenvectors, as for $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, and no change of field helps. In the second case we should ask for less. The next simplest shape after a diagonal matrix is an upper triangular one, and triangular matrices are almost as good for computation: their eigenvalues, trace and determinant are visible on the diagonal. This section proves that an operator has a triangular matrix exactly when its characteristic polynomial splits, so every complex matrix is similar to a triangular one. The proof is the induction on dimension that Section 1 promised: take one eigenvector, pass to the quotient, and repeat.

8.7.1 Triangular matrices and chains of invariant subspaces

As in Section 2, start with the wish and unwind it. Let $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ be a basis of V and $T \in \mathcal{L}(V)$. Column j of $[T]_{\mathcal{B}}$ holds the coordinates of $T\mathbf{v}_j$ (Definition 3.6.1). The matrix is upper triangular when every entry below the diagonal is 0 (Definition 0.9.5), that is, when $T\mathbf{v}_j$ is a combination of $\mathbf{v}_1, \dots, \mathbf{v}_j$ only. So the first basis vector must be sent to a multiple of itself, the second into the plane of the first two, and so on. In the language of Section 1, each of the growing spans must be invariant. Chapter 3 proved this as an exercise (Exercise (C1: Upper triangular matrices and invariant subspaces)), where the word *invariant* was introduced just for that problem; here it is again, with one more piece of information about the diagonal.

Theorem 8.7.1 — Triangular Matrices and Chains of Invariant Subspaces

Let V be finite-dimensional with basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$, let $T \in \mathcal{L}(V)$, and put $U_0 = \{\mathbf{0}\}$ and $U_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ for $1 \leq k \leq n$. The following are equivalent:

- (a) $[T]_{\mathcal{B}}$ is upper triangular;
- (b) $T\mathbf{v}_k \in U_k$ for every $k = 1, \dots, n$;
- (c) each U_k is T -invariant.

In that case, if $d_1, \dots, d_n$ are the diagonal entries of $[T]_{\mathcal{B}}$, then $(T - d_k \text{id}_V)(U_k) \subseteq U_{k-1}$ for every k .

Proof. Write $[T]_{\mathcal{B}} = (a_{ij})$, so that $T\mathbf{v}_j = \sum_i a_{ij}\mathbf{v}_i$ with unique coefficients (Definition 3.6.1, Theorem 1.5.1).

- (a) $\Leftrightarrow$ (b). If $a_{ij} = 0$ for $i > j$, then $T\mathbf{v}_j = \sum_{i \leq j} a_{ij}\mathbf{v}_i \in U_j$. Conversely, if $T\mathbf{v}_j \in U_j$, it has a representation using only $\mathbf{v}_1, \dots, \mathbf{v}_j$; by uniqueness, $a_{ij} = 0$ for $i > j$.
- (b) $\Rightarrow$ (c). For $j \leq k$, $T\mathbf{v}_j \in U_j \subseteq U_k$, so U_k is invariant by Lemma 8.1.1.
- (c) $\Rightarrow$ (b). $\mathbf{v}_k \in U_k$, so $T\mathbf{v}_k \in U_k$ by invariance.

Finally, suppose (a) holds and let $\mathbf{u} = \sum_{j \leq k} b_j \mathbf{v}_j \in U_k$. For $j < k$, $(T - d_k \text{id}_V)\mathbf{v}_j = T\mathbf{v}_j - d_k \mathbf{v}_j \in U_j \subseteq U_{k-1}$. For $j = k$, $(T - d_k \text{id}_V)\mathbf{v}_k = \sum_{i \leq k} a_{ik}\mathbf{v}_i - a_{kk}\mathbf{v}_k = \sum_{i < k} a_{ik}\mathbf{v}_i \in$

U_{k-1} , since $d_k = a_{kk}$. By linearity, $(T - d_k \text{id}_V)\mathbf{u} \in U_{k-1}$.

■

A chain $U_1 \subset U_2 \subset \cdots \subset U_n = V$ of subspaces with $\dim U_k = k$ is often called a **flag**, after the picture of a point on a line in a plane in space. The theorem says: an upper triangular matrix is the same thing as a flag of invariant subspaces, together with a basis adapted to it. The last statement says what the diagonal entry d_k measures. Modulo the smaller space U_{k-1} , the operator T acts on the new basis vector $\mathbf{v}_k$ as multiplication by d_k .

The operators for which such a basis exists deserve a name.

Definition 8.7.1 — Triangularizable

Let V be a finite-dimensional vector space over F . An operator $T \in \mathcal{L}(V)$ is **triangularizable over F** if there **exists** a basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}}$ is upper triangular. A matrix $\mathbf{A} \in M_n(F)$ is **triangularizable over F** if there **exist** an **invertible** $\mathbf{P} \in M_n(F)$ and an upper triangular $\mathbf{U} \in M_n(F)$ with $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$.

As for diagonalizability, $\mathbf{A}$ is triangularizable over F exactly when $T_{\mathbf{A}}$ is, by Theorem 3.7.4.

Examples.

- **Diagonalizable operators** are triangularizable, since a diagonal matrix is upper triangular. So are all operators on a space of dimension 1, and the zero and identity operators.
- **Differentiation.** For D on $\mathbb{R}[x]_{\leq n}$ in the basis $(1, x, \dots, x^n)$, the matrix is upper triangular with zero diagonal: $D(x^k) = kx^{k-1} \in \text{span}(1, \dots, x^k)$. The flag is $\mathbb{R}[x]_{\leq 0} \subset \mathbb{R}[x]_{\leq 1} \subset \cdots \subset \mathbb{R}[x]_{\leq n}$, a chain of invariant subspaces we met in Section 1. D is not diagonalizable (its only eigenvectors are the constants), but it is triangularizable.
- **Too few eigenvectors.** $\mathbf{J} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is already triangular, with flag $\text{span}(\mathbf{e}_1) \subset F^2$.

Non-example by minimal change. Reverse the basis for D on $\mathbb{R}[x]_{\leq 2}$: take $(x^2, x, 1)$. The spans are $\text{span}(x^2) \subset \text{span}(x^2, x) \subset \mathbb{R}[x]_{\leq 2}$, still a chain of subspaces of dimensions 1, 2, 3. What fails is invariance of the first one: $D(x^2) = 2x \notin \text{span}(x^2)$. Accordingly, the matrix in this basis is

$$\begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix},$$

which is lower triangular, not upper. Being triangularizable is a property of T ; being triangular is a property of T **and** a basis.

A triangularizable operator on $V \neq \{\mathbf{0}\}$ must have an invariant line $U_1 = \text{span}(\mathbf{v}_1)$, hence an eigenvector (Proposition 8.1.2). So the rotation R of $\mathbb{R}^2$ by a right angle, which has no invariant line (Example (A Rotation Has No Invariant Lines)), is **not** triangularizable over $\mathbb{R}$. An eigenvector to start the flag is exactly what is needed, and the next theorem shows that it is also enough, provided eigenvectors keep existing at every stage.

Quick check. Is $\begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix} \in M_2(\mathbb{R})$ triangularizable over $\mathbb{R}$? If so, find a basis in which its matrix is upper triangular.

Solution. Yes. We need a first basis vector that is an eigenvector. The matrix sends $\mathbf{e}_2$ to $(0, 2) = 2\mathbf{e}_2$, so start with $\mathbf{e}_2$, and complete with $\mathbf{e}_1$. In the basis $(\mathbf{e}_2, \mathbf{e}_1)$, the image of $\mathbf{e}_2$ is $2\mathbf{e}_2$ and the image of $\mathbf{e}_1$ is $(2, 1) = \mathbf{e}_2 + 2\mathbf{e}_1$, so the matrix is $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$, which is upper triangular. (The standard basis gives a lower triangular matrix: $\text{span}(\mathbf{e}_1)$ is not invariant.)

8.7.2 The Triangularization Theorem

If T has a triangular matrix, its characteristic polynomial is a product of linear factors, one for each diagonal entry. The converse is the real content: splitting of p_T is enough to build the flag.

Theorem 8.7.2 — Triangularization Theorem

Let V be a finite-dimensional vector space over F with $\dim V = n \geq 1$, and let $T \in \mathcal{L}(V)$. Then T is triangularizable over F if and only if p_T splits over F . The same holds for $\mathbf{A} \in M_n(F)$ and $p_{\mathbf{A}}$.

Idea. Take one eigenvector, extend, recurse. Since p_T splits, it has a root in F , so T has an eigenvector $\mathbf{v}_1$. Put $U = \text{span}(\mathbf{v}_1)$ and extend $\mathbf{v}_1$ to a basis. By Theorem 8.1.3 the matrix looks like

$$\begin{pmatrix} \lambda & * \\ 0 & [\bar{T}] \end{pmatrix},$$

where $\bar{T}$ is the induced operator on the smaller space V/U . The first column is already right. If $[\bar{T}]$ could be made triangular, the whole matrix would be. Now $\bar{T}$ lives on a space of dimension $n - 1$, and $p_T = (x - \lambda)p_{\bar{T}}$, so $p_{\bar{T}}$ splits too: this is the induction on dimension. The only technical point is that a basis of V/U lifts to vectors that, together with $\mathbf{v}_1$, form a basis of V .

Proof. ($\Rightarrow$) Let $[T]_{\mathcal{B}}$ be upper triangular with diagonal entries $d_1, \dots, d_n$. The matrix $x\mathbf{I}_n - [T]_{\mathcal{B}}$ is upper triangular with diagonal entries $x - d_i$, and Theorem 6.4.6 holds over $F[x]$ (the remark at the end of Chapter 6, §4). So $p_T = (x - d_1) \cdots (x - d_n)$ by Definition 6.10.2, which splits (Definition 5.4.2).

($\Leftarrow$) We prove, by induction on $n \geq 1$, that every operator on an n -dimensional space whose characteristic polynomial splits is triangularizable. For $n = 1$, every 1×1 matrix is upper triangular.

Let $n \geq 2$, and assume the statement for spaces of dimension $n - 1$. Since p_T is monic of degree n (Theorem 6.10.2) and splits, $p_T = (x - c_1)(x - c_2) \cdots (x - c_n)$ with $c_i \in F$. As $p_T(c_1) = 0$, Theorem 8.2.1 gives an eigenvector $\mathbf{v}_1$ with $T\mathbf{v}_1 = c_1\mathbf{v}_1$. Let $U = \text{span}(\mathbf{v}_1)$. It is T -invariant (Proposition 8.1.2), $\dim U = 1$, and $U \neq V$ since $n \geq 2$. Let $\bar{T}$ be the induced operator on V/U (Definition 8.1.2), a space of dimension $n - 1 \geq 1$ (Theorem 3.9.5).

$p_{\bar{T}}$ splits. The restriction $T|_U$ is $c_1 \text{id}_U$, whose matrix in the basis $(\mathbf{v}_1)$ is (c_1) , so $p_{T|_U} = x - c_1$. By Theorem 8.1.3 (b),

$$(x - c_1)(x - c_2) \cdots (x - c_n) = p_T = (x - c_1)p_{\bar{T}}.$$

Canceling the non-zero factor $x - c_1$ (Corollary 0.8.4 (2)) gives $p_{\bar{T}} = (x - c_2) \cdots (x - c_n)$, which splits.

Apply the induction hypothesis. There is a basis $\bar{\mathcal{C}} = (\mathbf{w}_2 + U, \dots, \mathbf{w}_n + U)$ of V/U with $[\bar{T}]_{\bar{\mathcal{C}}}$ upper triangular, for some $\mathbf{w}_2, \dots, \mathbf{w}_n \in V$.

Lift the basis. Let $\mathcal{B} = (\mathbf{v}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$. Suppose $a\mathbf{v}_1 + b_2\mathbf{w}_2 + \cdots + b_n\mathbf{w}_n = \mathbf{0}$. Passing to cosets, and using $\mathbf{v}_1 \in U$, gives $b_2(\mathbf{w}_2 + U) + \cdots + b_n(\mathbf{w}_n + U) = \mathbf{0} + U$ (Theorem 3.9.4). Since $\bar{\mathcal{C}}$ is a basis, all $b_i = 0$. Then $a\mathbf{v}_1 = \mathbf{0}$ with $\mathbf{v}_1 \neq \mathbf{0}$, so $a = 0$ (Theorem 1.1.9). Thus $\mathcal{B}$ is a linearly independent list of n vectors in V , hence a basis (Theorem 1.6.4).

The matrix. $\mathcal{B}$ extends the basis $(\mathbf{v}_1)$ of U , and the cosets of the added vectors form $\bar{\mathcal{C}}$. By Theorem 8.1.3 (b),

$$[T]_{\mathcal{B}} = \begin{pmatrix} c_1 & * \\ 0 & [\bar{T}]_{\bar{\mathcal{C}}} \end{pmatrix},$$

where the 0 is a column of $n - 1$ zeros. The block $[\bar{T}]_{\bar{\mathcal{C}}}$ is upper triangular, so $[T]_{\mathcal{B}}$ is upper triangular. This completes the induction.

For $\mathbf{A} \in M_n(F)$, apply the result to $T_{\mathbf{A}}$, using Theorem 3.7.4.

■

The proof is an algorithm, and its output is a flag. The first invariant subspace is the eigenline U ; the k -th is spanned by $\mathbf{v}_1$ and the lifts $\mathbf{w}_2, \dots, \mathbf{w}_k$ of the first $k - 1$ vectors of the flag for $\bar{T}$. The eigenvalue found at each stage is the next diagonal entry.

Over the complex numbers the hypothesis is automatic.

Corollary 8.7.3 — Complex Operators Are Triangularizable

Every operator on a finite-dimensional complex vector space $V \neq \{\mathbf{0}\}$ is triangularizable over $\mathbb{C}$. Every $\mathbf{A} \in M_n(\mathbb{C})$, $n \geq 1$, is similar to an upper triangular matrix in $M_n(\mathbb{C})$.

Proof. By Theorem 6.10.2, p_T is non-zero, and by Corollary 5.5.5 (a) it splits over $\mathbb{C}$. Apply Theorem 8.7.2.

■

So over $\mathbb{C}$ the question of Chapter 3 always has at least this answer: a basis exists in which the matrix has zeros below the diagonal. A real matrix can always be triangularized as a complex matrix, but perhaps only with complex entries.

Warning. **Triangularizing may require leaving the field, and the triangular form is far from unique.** The rotation $\mathbf{R} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $p_{\mathbf{R}} = x^2 + 1$, which does not split over $\mathbb{R}$, so no **real** invertible $\mathbf{P}$ makes $\mathbf{P}^{-1}\mathbf{R}\mathbf{P}$ triangular; over $\mathbb{C}$ it becomes $\text{diag}(i, -i)$. And a triangularizable matrix has many triangular forms. $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ and $\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ are both triangular forms of the same operator (the first has distinct eigenvalues 1, 2, so it is diagonalizable, and similar to the second). The off-diagonal entries change with the flag and the basis, and even the order of the diagonal entries is not fixed. Only

the diagonal entries, as a list with repetitions, are determined, as the next theorem shows.

8.7.3 What the diagonal records

A diagonalization displays the eigenvalues. A triangularization does too, with their algebraic multiplicities.

Theorem 8.7.4 — The Diagonal of a Triangular Form

Let V be finite-dimensional with $\dim V = n \geq 1$, $T \in \mathcal{L}(V)$, and let $\mathcal{B}$ be a basis with $[T]_{\mathcal{B}}$ upper triangular, with diagonal entries $d_1, \dots, d_n$. Then:

- (a) $p_T = (x - d_1)(x - d_2) \cdots (x - d_n)$;
- (b) $\text{spec}(T) = \{d_1, \dots, d_n\}$, and each eigenvalue λ occurs among $d_1, \dots, d_n$ exactly $a_T(\lambda)$ times;
- (c) $\text{tr } T = d_1 + \cdots + d_n$ and $\det T = d_1 \cdots d_n$.

In particular, any two triangular forms of T have the same diagonal entries, up to order.

Proof.

- (a) was shown in the proof of Theorem 8.7.2, ($\Rightarrow$).
- (b) By Theorem 8.2.1, the eigenvalues are the roots of p_T in F , and λ is a root of $\prod_i (x - d_i)$ exactly when $\prod_i (\lambda - d_i) = 0$, that is, when $\lambda = d_i$ for some i (Theorem 0.7.1). By Theorem 5.4.2, $a_T(\lambda) = \text{mult}_{\lambda}(p_T) = \sum_i \text{mult}_{\lambda}(x - d_i)$. Each term is 1 if $d_i = \lambda$ and 0 otherwise, by Theorem 5.1.3 (b), since $x - d_i$ is not divisible by $(x - \lambda)^2$ for degree reasons. So the sum counts the indices with $d_i = \lambda$.
- (c) By Definition 3.10.2, $\text{tr } T = \text{tr}[T]_{\mathcal{B}} = \sum_i d_i$. By Definition 6.5.1 and Theorem 6.4.6, $\det T = \det[T]_{\mathcal{B}} = \prod_i d_i$.

The final statement follows from (b): the number of times λ occurs on the diagonal is $a_T(\lambda)$, which does not depend on the basis. ■

Part (c) recovers Theorem 8.3.3. In Section 3 it came from comparing coefficients of p_T ; here it is read off a matrix. For instance, a complex 3×3 matrix with $p_{\mathbf{A}} = (x - 1)^2(x + 4)$ is similar to a triangular matrix with diagonal 1, 1, -4 in some order, so $\text{tr } \mathbf{A} = -2$ and $\det \mathbf{A} = -4$, however complicated $\mathbf{A}$ looks.

The theorem does **not** say that a triangular form with **distinct** diagonal entries is itself diagonal: $\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$ is a counterexample. What does follow is that T is then diagonalizable (Corollary 8.4.2), so some **other** basis makes its matrix diagonal. And an operator with repeated diagonal entries may or may not be diagonalizable: compare $\mathbf{I}_2$ with $\mathbf{J}$.

8.7.4 Triangularizing a matrix in practice

For a matrix $\mathbf{A} \in M_n(F)$ whose characteristic polynomial splits, the proof of [Theorem 8.7.2](#) becomes a procedure.

1. Find an eigenvalue λ_1 and an eigenvector $\mathbf{v}_1$.
2. Extend $\mathbf{v}_1$ to a basis $(\mathbf{v}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ of F^n , choosing the $\mathbf{w}_i$ as simple as possible, for example standard basis vectors.
3. Compute the matrix of the induced operator on $F^n / \text{span}(\mathbf{v}_1)$ in the basis $(\mathbf{w}_2 + U, \dots, \mathbf{w}_n + U)$. Its entries are the coordinates of $\mathbf{A}\mathbf{w}_j$ along $\mathbf{w}_2, \dots, \mathbf{w}_n$, ignoring the $\mathbf{v}_1$ -coordinate.
4. Triangularize this $(n-1) \times (n-1)$ matrix in the same way, and lift each coset back to its representative.

Example. (Triangularizing a Matrix with a Single Eigenvector Line)

Let

$$\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 1 & -1 & 2 \end{pmatrix} \in M_3(\mathbb{R}).$$

Show that $\mathbf{A}$ is not diagonalizable but is triangularizable over $\mathbb{R}$, and find an invertible $\mathbf{P}$ with $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ upper triangular.

Solution. The characteristic polynomial. Expanding along the first row (Theorem 6.6.1 over $\mathbb{R}[x]$),

$$\begin{aligned} p_{\mathbf{A}} &= \det \begin{pmatrix} x-2 & 0 & -1 \\ 0 & x-2 & -1 \\ -1 & 1 & x-2 \end{pmatrix} \\ &= (x-2)[(x-2)^2 + 1] + (-1) \det \begin{pmatrix} 0 & x-2 \\ -1 & 1 \end{pmatrix}. \end{aligned}$$

The last determinant is $0 \cdot 1 - (x-2)(-1) = x-2$, so $p_{\mathbf{A}} = (x-2)[(x-2)^2 + 1 - 1] = (x-2)^3$. It splits, so $\mathbf{A}$ is triangularizable by [Theorem 8.7.2](#).

Eigenvectors. $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$ has rank 2, so $g(2) = 1 < 3 = a(2)$, and $\mathbf{A}$ is not diagonalizable (Theorem 8.4.1 (e)). The system gives $z = 0$ and $x = y$, so $E_2(\mathbf{A}) = \text{span}(\mathbf{v}_1)$ with $\mathbf{v}_1 = (1, 1, 0)$: all eigenvectors lie on one line.

Step 1. Let $U = \text{span}(\mathbf{v}_1)$, and extend to the basis $(\mathbf{v}_1, \mathbf{e}_2, \mathbf{e}_3)$ of $\mathbb{R}^3$ (the matrix with these columns has determinant 1).

Step 2: the induced operator. Express the images of $\mathbf{e}_2$ and $\mathbf{e}_3$ in the basis:

$$\mathbf{A}\mathbf{e}_2 = (0, 2, -1) = 0\mathbf{v}_1 + 2\mathbf{e}_2 - \mathbf{e}_3, \quad \mathbf{A}\mathbf{e}_3 = (1, 1, 2) = 1\mathbf{v}_1 + 0\mathbf{e}_2 + 2\mathbf{e}_3.$$

Dropping the $\mathbf{v}_1$ -coordinates, the induced operator $\bar{\mathbf{A}}$ on $\mathbb{R}^3/U$ has, in the basis $(\mathbf{e}_2 + U, \mathbf{e}_3 + U)$, the matrix

$$[\bar{\mathbf{A}}] = \begin{pmatrix} 2 & 0 \\ -1 & 2 \end{pmatrix},$$

with $p_{\bar{\mathbf{A}}} = (x-2)^2$, as the proof predicts.

Step 3: triangularize $\bar{\mathbf{A}}$. Its second column says $\bar{\mathbf{A}}(\mathbf{e}_3 + U) = 2(\mathbf{e}_3 + U)$, so $\mathbf{e}_3 + U$ is an eigenvector. In the basis $(\mathbf{e}_3 + U, \mathbf{e}_2 + U)$, $\bar{\mathbf{A}}(\mathbf{e}_2 + U) = -(\mathbf{e}_3 + U) + 2(\mathbf{e}_2 + U)$, and the matrix becomes $\begin{pmatrix} 2 & -1 \\ 0 & 2 \end{pmatrix}$.

Step 4: lift. The basis $\mathcal{B} = (\mathbf{v}_1, \mathbf{e}_3, \mathbf{e}_2)$ of $\mathbb{R}^3$ gives

$$\mathbf{P} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Check. $\mathbf{A}\mathbf{v}_1 = (2, 2, 0) = 2\mathbf{v}_1$; $\mathbf{A}\mathbf{e}_3 = (1, 1, 2) = \mathbf{v}_1 + 2\mathbf{e}_3$; $\mathbf{A}\mathbf{e}_2 = (0, 2, -1) = -\mathbf{e}_3 + 2\mathbf{e}_2$. These are the columns $(2, 0, 0)$, $(1, 2, 0)$, $(0, -1, 2)$ of the triangular matrix, read in the basis $\mathcal{B}$. Also $\det \mathbf{P} = -1 \neq 0$.

The flag is $\text{span}((1, 1, 0)) \subset \{(a, a, b) : a, b \in \mathbb{R}\} \subset \mathbb{R}^3$. One can check the plane directly: $\mathbf{A}(a, a, b) = (2a + b, 2a + b, 2b)$ lies in it again. The diagonal is 2, 2, 2, as [Theorem 8.7.4](#) requires.

Different choices in Step 2 give different triangular matrices, with the same diagonal. Chapter 9 shows how to choose the basis so that the entries above the diagonal are as simple as possible (the Jordan form). Chapter 11 shows that over $\mathbb{C}$ with an inner product, the basis can even be chosen orthonormal (Schur's theorem).

8.7.5 Exercises

8.7.5.1 A. Check your understanding

Exercise A1.

- Define what it means for $\mathbf{A} \in M_n(F)$ to be triangularizable over F .
- State the Triangularization Theorem.
- For a basis $(\mathbf{v}_1, \dots, \mathbf{v}_n)$, which subspaces must be invariant for $[T]_{\mathcal{B}}$ to be upper triangular?
- True or false: every $\mathbf{A} \in M_n(\mathbb{R})$ is triangularizable over $\mathbb{R}$. Justify your answer.
- True or false: every triangularizable matrix is diagonalizable. Justify your answer.
- Let $\mathbf{A} \in M_4(\mathbb{C})$ have $p_{\mathbf{A}} = (x - 3)^3(x + 1)$. What are the diagonal entries of any upper triangular matrix similar to $\mathbf{A}$, and what are $\text{tr } \mathbf{A}$ and $\det \mathbf{A}$?

Solution.

- There are an invertible $\mathbf{P} \in M_n(F)$ and an upper triangular $\mathbf{U} \in M_n(F)$ with $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$ ([Definition 8.7.1](#)).
- An operator on a space of dimension $n \geq 1$ over F is triangularizable over F if and only if its characteristic polynomial splits over F ([Theorem 8.7.2](#)).
- All the spans $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$, $1 \leq k \leq n$ ([Theorem 8.7.1](#)).
- False. The rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has $p = x^2 + 1$, which does not split over $\mathbb{R}$.

- (e) False. $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ is triangular but not diagonalizable.
- (f) By [Theorem 8.7.4](#), the diagonal is 3, 3, 3, -1 in some order, so $\operatorname{tr} \mathbf{A} = 8$ and $\det \mathbf{A} = -27$.

8.7.5.2 B. Practice

Exercise B1: Triangularizing matrices.

For each real matrix, find an invertible $\mathbf{P}$ such that $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$ is upper triangular, and write down $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$.

- (a) $\mathbf{A} = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix}$.
- (b) $\mathbf{A} = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix}$, with $p_{\mathbf{A}} = (x-1)(x-3)^2$ (Example (A 3×3 Characteristic Polynomial)).

Solution.

- (a) $\operatorname{tr} \mathbf{A} = 4$, $\det \mathbf{A} = 4$, so $p_{\mathbf{A}} = (x-2)^2$, which splits. $\mathbf{A} - 2\mathbf{I} = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ gives the eigenvector $\mathbf{v}_1 = (1, -1)$. Extend with $\mathbf{e}_2$: $\mathbf{A}\mathbf{e}_2 = (1, 1) = \mathbf{v}_1 + 2\mathbf{e}_2$. So with

$$\mathbf{P} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

(In size 2, one eigenvector already fixes the flag.)

- (b) $\mathbf{A}\mathbf{e}_2 = (0, 3, 0) = 3\mathbf{e}_2$, so $\mathbf{v}_1 = \mathbf{e}_2$ is an eigenvector for 3. Let $U = \operatorname{span}(\mathbf{e}_2)$ and extend to $(\mathbf{e}_2, \mathbf{e}_1, \mathbf{e}_3)$. Then $\mathbf{A}\mathbf{e}_1 = (2, 1, 1) = \mathbf{e}_2 + 2\mathbf{e}_1 + \mathbf{e}_3$ and $\mathbf{A}\mathbf{e}_3 = (1, 1, 2) = \mathbf{e}_2 + \mathbf{e}_1 + 2\mathbf{e}_3$, so the induced operator has matrix $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ in the basis $(\mathbf{e}_1 + U, \mathbf{e}_3 + U)$. Its characteristic polynomial is $x^2 - 4x + 3 = (x-3)(x-1)$, with eigenvectors $(1, 1)$ for 3 and $(1, -1)$ for 1, that is, $\mathbf{e}_1 + \mathbf{e}_3 + U$ and $\mathbf{e}_1 - \mathbf{e}_3 + U$. Lifting gives the basis $(\mathbf{e}_2, \mathbf{e}_1 + \mathbf{e}_3, \mathbf{e}_1 - \mathbf{e}_3)$. Now [

$$\mathbf{A}(\mathbf{e}_1 + \mathbf{e}_3) = (3, 2, 3) = 2\mathbf{e}_2 + 3(\mathbf{e}_1 + \mathbf{e}_3),$$

$$\mathbf{A}(\mathbf{e}_1 - \mathbf{e}_3) = (1, 0, -1) = \mathbf{e}_1 - \mathbf{e}_3,$$

] so

$$\mathbf{P} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & -1 \end{pmatrix}, \quad \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 3 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

with $\det \mathbf{P} = 2 \neq 0$ (expand along the second row). The diagonal 3, 3, 1 agrees with $p_{\mathbf{A}}$.

Exercise B2: Nilpotent matrices are strictly triangular.

Let $\mathbf{A} \in M_n(\mathbb{C})$, $n \geq 1$, with $\mathbf{A}^k = 0$ for some $k \geq 1$. Prove that $\mathbf{A}$ is similar to a **strictly** upper triangular matrix, that is, an upper triangular matrix with zero diagonal.

Solution. By Corollary 8.7.3, $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$ is upper triangular for some invertible $\mathbf{P}$. By Theorem 8.7.4 (b), each diagonal entry of $\mathbf{U}$ is an eigenvalue of $\mathbf{A}$. By Exercise (C2: Nilpotent operators) (a), the only eigenvalue of a nilpotent operator on a non-zero space is 0. So every diagonal entry of $\mathbf{U}$ is 0, and $\mathbf{U}$ is strictly upper triangular. (The statement holds over every field, with a different proof: use the chain $\ker \mathbf{A} \subseteq \ker \mathbf{A}^2 \subseteq \cdots \subseteq \ker \mathbf{A}^k = F^n$ to build the basis.)

Exercise B3: Eigenvalues of powers from a triangular form.

Let $\mathbf{A} \in M_n(\mathbb{C})$, $n \geq 1$, and let $\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{U}$ be upper triangular with diagonal entries $d_1, \dots, d_n$.

- (a) Show that the product of two upper triangular matrices is upper triangular, with diagonal entries the products of the corresponding diagonal entries.
- (b) Deduce that $p_{\mathbf{A}^k} = (x - d_1^k) \cdots (x - d_n^k)$ and $\text{tr}(\mathbf{A}^k) = d_1^k + \cdots + d_n^k$ for every $k \geq 1$.
- (c) Suppose $p_{\mathbf{A}} = (x - 1)(x + 1)(x - 2)$. Find $p_{\mathbf{A}^2}$, $\text{tr} \mathbf{A}^2$ and $\det \mathbf{A}^2$.

Solution.

- (a) Let $\mathbf{U}, \mathbf{U}'$ be upper triangular. The (i, j) -entry of $\mathbf{U}\mathbf{U}'$ is $\sum_l u_{il}u'_{lj}$. A term can be non-zero only if $u_{il} \neq 0$ and $u'_{lj} \neq 0$, which forces $i \leq l \leq j$. So the entry is 0 when $i > j$, and for $i = j$ only $l = i$ survives, giving $u_{ii}u'_{ii}$.
- (b) $\mathbf{P}^{-1}\mathbf{A}^k\mathbf{P} = (\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^k = \mathbf{U}^k$, by induction on k . By (a) and induction, $\mathbf{U}^k$ is upper triangular with diagonal entries d_i^k . By Theorem 8.7.4 (a) and (c), applied to $T_{\mathbf{A}^k}$ in the basis given by the columns of $\mathbf{P}$, $p_{\mathbf{A}^k} = \prod_i (x - d_i^k)$ and $\text{tr} \mathbf{A}^k = \sum_i d_i^k$.
- (c) By Theorem 8.7.4, the diagonal of $\mathbf{U}$ is 1, -1, 2 in some order. By (b), $p_{\mathbf{A}^2} = (x - 1)(x - 1)(x - 4) = (x - 1)^2(x - 4)$, $\text{tr} \mathbf{A}^2 = 1 + 1 + 4 = 6$, and $\det \mathbf{A}^2 = 1 \cdot 1 \cdot 4 = 4$. The last agrees with $\det \mathbf{A}^2 = (\det \mathbf{A})^2 = (-2)^2$.

8.7.5.3 C. Going deeper

Exercise C1: A second proof of Cayley–Hamilton.

Let V be finite-dimensional with $\dim V = n \geq 1$, and let $T \in \mathcal{L}(V)$ have a basis $\mathcal{B} = (\mathbf{v}_1, \dots, \mathbf{v}_n)$ with $[T]_{\mathcal{B}}$ upper triangular, with diagonal entries $d_1, \dots, d_n$. Let $U_0 = \{\mathbf{0}\}$ and $U_k = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$.

- (a) Let $S_k = (T - d_1 \text{id}_V)(T - d_2 \text{id}_V) \cdots (T - d_k \text{id}_V)$. Prove by induction on k that $S_k(U_k) = \{\mathbf{0}\}$ for $1 \leq k \leq n$.
- (b) Deduce that $p_T(T) = 0$, without using the adjugate.
- (c) Deduce the Cayley–Hamilton Theorem for every $\mathbf{A} \in M_n(\mathbb{C})$, and then for every $\mathbf{A} \in M_n(\mathbb{R})$.
- (d) Why does this argument, on its own, not prove the theorem for $\mathbf{A} \in M_2(\mathbb{F}_2)$ with $p_{\mathbf{A}} = x^2 + x + 1$?

Hint: for (a), use the last statement of Theorem 8.7.1.

Solution.

- (a) By [Theorem 8.7.1](#), each U_k is invariant and $(T - d_k \text{id}_V)(U_k) \subseteq U_{k-1}$. For $k = 1$, $S_1(U_1) = (T - d_1 \text{id}_V)(U_1) \subseteq U_0 = \{\mathbf{0}\}$. Suppose $S_{k-1}(U_{k-1}) = \{\mathbf{0}\}$ for some $k \geq 2$. Since $S_k = S_{k-1}(T - d_k \text{id}_V)$, for $\mathbf{u} \in U_k$ we get $S_k \mathbf{u} = S_{k-1}((T - d_k \text{id}_V)\mathbf{u}) \in S_{k-1}(U_{k-1}) = \{\mathbf{0}\}$.
- (b) By [Theorem 8.7.4](#) (a), $p_T = \prod_{i=1}^n (x - d_i)$, so $p_T(T) = S_n$ by Theorem 5.7.1 (b). By (a), S_n kills $U_n = V$, so $p_T(T) = 0$.
- (c) Let $\mathbf{A} \in M_n(\mathbb{C})$. By [Corollary 8.7.3](#), $T_{\mathbf{A}}$ has a basis in which its matrix is upper triangular, and (b) gives $p_{\mathbf{A}}(T_{\mathbf{A}}) = 0$, so $p_{\mathbf{A}}(\mathbf{A}) = 0$ by Corollary 3.6.3. For $\mathbf{A} \in M_n(\mathbb{R})$, regard $\mathbf{A}$ as an element of $M_n(\mathbb{C})$. The Leibniz formula gives the same polynomial $p_{\mathbf{A}}$ with real coefficients, and $p_{\mathbf{A}}(\mathbf{A})$ is computed from the same entries by the same sums and products; it is 0 by the complex case.
- (d) Over $\mathbb{F}_2$, $x^2 + x + 1$ has no root (its values at 0 and 1 are both 1), so it does not split, and the matrix is not triangularizable over $\mathbb{F}_2$ by [Theorem 8.7.2](#). To run the argument we would need a larger field containing $\mathbb{F}_2$ in which the polynomial splits, and we have not constructed one. The adjugate proof of Theorem 8.6.1 needs no such field. (For instance $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ has this characteristic polynomial, and indeed $\mathbf{A}^2 + \mathbf{A} + \mathbf{I} = 0$ over $\mathbb{F}_2$.)

Exercise C2: Real operators have small invariant subspaces.

Let $\mathbf{A} \in M_n(\mathbb{R})$, $n \geq 1$, regarded also as a complex matrix.

- (a) Let $\lambda = a + bi$ ($a, b \in \mathbb{R}$) be a complex eigenvalue of $\mathbf{A}$ with eigenvector $\mathbf{z} = \mathbf{x} + i\mathbf{y}$, where $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Show that $\mathbf{A}\mathbf{x} = a\mathbf{x} - b\mathbf{y}$ and $\mathbf{A}\mathbf{y} = b\mathbf{x} + a\mathbf{y}$.
- (b) Deduce that $\text{span}(\mathbf{x}, \mathbf{y}) \subseteq \mathbb{R}^n$ is $\mathbf{A}$ -invariant, and has dimension 1 or 2.
- (c) Deduce that every operator on a real vector space V with $1 \leq \dim V < \infty$ has an invariant subspace of dimension 1 or 2.
- (d) For $\mathbf{C} = \begin{pmatrix} 0 & 0 & 2 \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{pmatrix}$, check that $\mathbf{z} = (0, -2, 1) + i(-2, 1, 0)$ is an eigenvector for i , and find the invariant plane of (b).

Solution.

- (a) The eigenvalue exists by Theorem 8.2.4. Since $\mathbf{A}$ has real entries, $\mathbf{A}\mathbf{z} = \mathbf{A}\mathbf{x} + i\mathbf{A}\mathbf{y}$ with $\mathbf{A}\mathbf{x}, \mathbf{A}\mathbf{y} \in \mathbb{R}^n$. On the other hand, $\lambda\mathbf{z} = (a + bi)(\mathbf{x} + i\mathbf{y}) = (a\mathbf{x} - b\mathbf{y}) + i(b\mathbf{x} + a\mathbf{y})$. A vector of $\mathbb{C}^n$ determines its real and imaginary parts entry by entry, so $\mathbf{A}\mathbf{x} = a\mathbf{x} - b\mathbf{y}$ and $\mathbf{A}\mathbf{y} = b\mathbf{x} + a\mathbf{y}$.
- (b) By (a), $\mathbf{A}\mathbf{x}$ and $\mathbf{A}\mathbf{y}$ lie in $W = \text{span}(\mathbf{x}, \mathbf{y})$, so W is invariant by Lemma 8.1.1. Since $\mathbf{z} \neq \mathbf{0}$, at least one of $\mathbf{x}, \mathbf{y}$ is non-zero, so $\dim W \geq 1$, and $\dim W \leq 2$ since W is spanned by two vectors.
- (c) Choose a basis $\mathcal{B}$ of V , let $\mathbf{A} = [T]_{\mathcal{B}} \in M_n(\mathbb{R})$, and let W be as in (b). Let $\Phi: V \rightarrow \mathbb{R}^n$ be the coordinate isomorphism $\mathbf{v} \mapsto [\mathbf{v}]_{\mathcal{B}}$ (Corollary 3.5.5). Then $\Phi^{-1}(W)$ is a subspace of the same dimension as W , and for $\mathbf{v} \in \Phi^{-1}(W)$, $\Phi(T\mathbf{v}) = \mathbf{A}\Phi(\mathbf{v}) \in W$ (Theorem 3.6.1), so $T\mathbf{v} \in \Phi^{-1}(W)$.
- (d) With $\mathbf{x} = (0, -2, 1)$ and $\mathbf{y} = (-2, 1, 0)$: $\mathbf{C}\mathbf{x} = (2, -1, 0) = -\mathbf{y}$ and $\mathbf{C}\mathbf{y} =$

$(0, -2, 1) = \mathbf{x}$. So $\mathbf{C}\mathbf{z} = \mathbf{C}\mathbf{x} + i\mathbf{C}\mathbf{y} = -\mathbf{y} + i\mathbf{x} = i(\mathbf{x} + i\mathbf{y}) = i\mathbf{z}$, and $\mathbf{z}$ is an eigenvector for i (here $a = 0$, $b = 1$, matching (a)). The invariant plane is $\text{span}((0, -2, 1), (-2, 1, 0))$, which is the plane $x + 2y + 4z = 0$. Over $\mathbb{R}$, $\mathbf{C}$ also has the eigenvalue 2 with eigenvector $(1, 0, 1)$, an invariant line.

Exercise C3: Traces of powers detect nilpotency.

Let $\mathbf{C} \in M_n(\mathbb{C})$, $n \geq 1$, with $\text{tr}(\mathbf{C}^k) = 0$ for $k = 1, \dots, n$. The goal is to prove that $\mathbf{C}^n = 0$.

- (a) Let $d_1, \dots, d_n$ be the diagonal entries of a triangular form of $\mathbf{C}$. Show that $d_1^k + \dots + d_n^k = 0$ for $k = 1, \dots, n$.
- (b) Suppose some $d_i \neq 0$. Let $\mu_1, \dots, \mu_r$ ($r \geq 1$) be the distinct non-zero values among the d_i , and let $m_j \geq 1$ be the number of indices i with $d_i = \mu_j$. Show that $\sum_{j=1}^r m_j \mu_j^k = 0$ for $k = 1, \dots, r$.
- (c) Show that the $r \times r$ matrix with (k, j) -entry μ_j^k is invertible, and derive a contradiction from (b).
- (d) Conclude that every $d_i = 0$, that $p_{\mathbf{C}} = x^n$, and that $\mathbf{C}^n = 0$.
- (e) Show that the statement fails over $\mathbb{F}_2$.

Hint: for (c), factor the matrix as a transposed Vandermonde matrix times a diagonal matrix.

Solution.

- (a) By [Corollary 8.7.3](#) a triangular form exists, and by [Exercise \(B3: Eigenvalues of powers from a triangular form\)](#) (b), $\text{tr } \mathbf{C}^k = \sum_i d_i^k$, which is 0 by hypothesis.
- (b) Indices with $d_i = 0$ contribute nothing to $\sum_i d_i^k$ for $k \geq 1$. Grouping the other indices by their value, $\sum_i d_i^k = \sum_j m_j \mu_j^k$. Since $r \leq n$, (a) applies for $k = 1, \dots, r$.
- (c) Let $\mathbf{M}$ be the matrix with (k, j) -entry μ_j^k . The Vandermonde matrix $\mathbf{V} = \mathbf{V}(\mu_1, \dots, \mu_r)$ has (j, k) -entry μ_j^{k-1} , so $\mathbf{V}^\top$ has (k, j) -entry μ_j^{k-1} , and multiplying column j by μ_j gives $\mathbf{M} = \mathbf{V}^\top \text{diag}(\mu_1, \dots, \mu_r)$. Since the μ_j are distinct, $\mathbf{V}$ is invertible ([Corollary 6.8.2](#)), hence so is $\mathbf{V}^\top$ ([Theorem 6.4.5](#)); since the μ_j are non-zero, the diagonal matrix is invertible. So $\mathbf{M}$ is invertible. By (b), $\mathbf{M}(m_1, \dots, m_r) = \mathbf{0}$, so $(m_1, \dots, m_r) = \mathbf{0}$ in $\mathbb{C}^r$. But each m_j is a positive integer, which is non-zero in $\mathbb{C}$. This contradiction shows that no d_i is non-zero.
- (d) By (c), all $d_i = 0$, so $p_{\mathbf{C}} = x^n$ by [Theorem 8.7.4](#) (a). By [Theorem 8.6.1](#), $\mathbf{C}^n = p_{\mathbf{C}}(\mathbf{C}) = 0$.
- (e) Over $\mathbb{F}_2$, $\mathbf{C} = \mathbf{I}_2$ has $\text{tr } \mathbf{C}^k = \text{tr } \mathbf{I}_2 = 1 + 1 = 0$ for every k , but $\mathbf{C}^k = \mathbf{I}_2 \neq 0$. The step that fails is (c): the positive integer $m_1 = 2$ is 0 in $\mathbb{F}_2$. Compare [Exercise \(C2: When \$\mathbf{A}\$ commutes with \$\[\mathbf{A}, \mathbf{B}\]\$ \)](#), where in characteristic 0 the same conclusion was reached for a commutator by a route that avoids eigenvalues.